SHUBHAM KISHOR VISPUTE

Embedded Firmware Engineer | Embedded Linux | Yocto | BSP | OpenBMC | C/C++ | Device Drivers
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PROFESSIONAL SUMMARY

Embedded Firmware / Embedded Linux Engineer with 6+ years of experience in Embedded C/C++, Embedded Linux, Yocto/OpenEmbedded and Linux device drivers on ARM platforms. Hands-on work on OpenBMC firmware for AST2600-based BMC platforms (Yocto image customization, D-Bus services, bmcweb/Redfish) and on NetSure power-controller firmware (custom Linux driver, Modbus, SNMP, relay and digital I/O integration). Experienced in hardware/software integration over CAN, I2C, SPI, UART, RS-485, Modbus and Ethernet-based protocols, with system-level debugging using GDB, Linux logs, Vehicle Spy and Wireshark. Working knowledge of kernel configuration/build, U-Boot and cross-compilation; conceptual understanding of embedded security. Works in Agile/Scrum teams and uses Cursor and GitHub Copilot to speed up development.

CORE TECHNICAL SKILLS

Programming: Embedded C, C++ (OOP, STL), Python, Shell Scripting, Data Structures, Pointers, Memory Management
Embedded Linux / BSP: Embedded Linux, Linux Device Drivers, Kernel Configuration & Build, U-Boot, Cross-compilation, systemd, Raspberry Pi
Yocto / OpenEmbedded: Yocto Project, OpenEmbedded, BitBake, Poky, Custom Layers, Recipes, Image Customization, QEMU
OpenBMC / BMC: OpenBMC, BMC Firmware, AST2600, D-Bus, bmcweb, Redfish, Entity Manager, WebUI, REST APIs, Sensor/Inventory Management Concepts
Protocols / Interfaces: CAN, UDS, LIN, I2C, SPI, UART, RS-485, Ethernet, Modbus, SNMP, TCP/IP, UDP, MQTT, HTTP/HTTPS; GPIO, PWM, Digital I/O, Relay Outputs
RTOS / Microcontrollers: FreeRTOS and RTOS Concepts, Interrupts, Timing, Concurrency, ARM, STM32, ESP32, ATmega32, Bare-metal Concepts
Debugging: GDB, Backtrace/Core Dumps, dmesg, strace, Linux Logs, Wireshark, Vehicle Spy, BusMaster, JTAG/SWD (concepts)
Tools / Build: Git, GitLab, QEMU, curl, Keil, STM32CubeIDE, MPLAB, Atmel Studio, Visual Studio, PyCharm, EasyEDA
AI Tools / Methodology: Cursor, GitHub Copilot (code generation, refactoring, debugging support, documentation); Agile, Scrum

EMBEDDED SECURITY

Hands-on: Secure HTTPS (TLS) based remote diagnostics and health checks using curl on the NetSure controller
Knowledge / Concepts: Secure Boot, Secure OTA/Firmware Update, Digital Signatures, AES, RSA, SHA-256, PKI and Digital Certificates, TLS Handshake, Key Management, Secure Coding Basics

PROFESSIONAL EXPERIENCE

Vertiv Energy Pvt. Ltd. | Embedded Firmware Engineer **Jun 2023 – Present**
Pune, Maharashtra

- Develop and customize OpenBMC firmware for AST2600-based ARM BMC platforms using Yocto/OpenEmbedded, building and validating images in QEMU ARM emulation for power-system controller applications.
- Implement and evaluate the OpenBMC management data path from hardware/controller data through Linux/BMC services and D-Bus to bmcweb/Redfish, exposing rectifier voltage, current, alarms and status; integrate WebUI components for system-management functionality.
- Developed a custom Linux device driver for communication between the IB5 interface board and the NetSure controller; integrated dual IB2 boards with relay outputs, digital inputs and temperature sensors.
- Designed and implemented Dual Generator Control supporting DG1, DG2 and Both modes in the NetSure controller firmware.
- Developed Modbus register mappings for lithium battery telemetry and control, and SNMP-based Fast Rectifier Turn-On/Turn-Off.
- Integrated curl-based secure-HTTPS remote diagnostics and health checks; debug embedded Linux and firmware issues using GDB, Linux logs and system diagnostics.
- Use AI-assisted development tools (Cursor, GitHub Copilot) for code generation, refactoring, debugging support and documentation, with all output reviewed and validated on target/QEMU.

- Work in an Agile/Scrum team, taking part in sprint planning, daily stand-ups and retrospectives to deliver firmware features in iterative sprints.

Intrepid Control Systems Pvt. Ltd. | Embedded Software Engineer **Sep 2022 – Jun 2023**
Pune, Maharashtra

- Worked on a Wireless Battery Management System (BMS) for automotive applications, supporting embedded software development and validation.
- Developed and modified APIs using C, C++ and Python based on project requirements.
- Investigated software and communication issues using Vehicle Spy and performed debugging/testing of automotive embedded devices and interfaces.

Guardian Media & Entertainment Pvt. Ltd. | R&D Embedded Engineer **Oct 2020 – Sep 2022**
Pune, Maharashtra

- Developed a Python Flask monitoring interface for show status, video-player health and operational logs.
- Designed PyQt5 GUI applications and Raspberry Pi-based video playback systems; integrated DMX lighting, GPIO and UDP trigger systems for synchronized multimedia applications.
- Developed multi-player video, Arduino-controlled audio and sensor-triggered systems across entertainment installations.

Intent Technologies Pvt. Ltd. | R&D Embedded Engineer **Oct 2019 – Oct 2020**
Ravet, Pune

- Designed PCBs using EasyEDA and developed ATmega32 firmware for headlamp, tail-lamp, fog-lamp and stop-lamp sequencing.
- Implemented RF remote-control functionality and PWM frequency-generation functionality for water-conditioning applications.

KEY PROJECTS

Design and Implementation of an Embedded Baseboard Management System Controller Using OpenBMC on Raspberry Pi 4
Raspberry Pi 4 | OpenBMC | Yocto | D-Bus | bmcweb | Redfish | Entity Manager | C++ | systemd

- Built an OpenBMC image for Raspberry Pi 4 using Yocto/OpenEmbedded (custom layers, recipe integration, kernel configuration) and ran BMC services under systemd.
- Implemented a rectifier-monitoring D-Bus service in C++ for power/rectifier hardware monitoring, integrated with Entity Manager.
- Exposed monitored data through bmcweb as Redfish and REST APIs. Architecture: hardware monitoring → D-Bus service → bmcweb → Redfish/REST client.

IoT-Based Automated Pet Feeder System

- ESP32-based automated pet feeder with scheduled dispensing through a mobile app using MQTT; integrated RTC, servo motor and OLED display.

Greenhouse Monitoring System

- ESP32-based monitoring system for temperature, humidity and soil moisture with web dashboard, alerts and MQTT/HTTP control.

EDUCATION

M.Tech in Embedded Systems - WILP, BITS Pilani | June 2024 – June 2026 | 7.2 CGPA
Diploma in Embedded System Design - BICARD, Pune (TCIL-IT, Delhi) | 2018 – 2019
B.Tech in Electronics & Telecommunication Engineering - Dr. Babasaheb Ambedkar Technological University, Lonere | 2015 – 2018 | 72.93%
Diploma in Electronics & Communication Engineering - S.S.V.P.S. B.S. Deore Polytechnic, Dhule | 2012 – 2015 | 81.03%

GITHUB PORTFOLIO AND IMPORTANT LINKS

- IoT-Based Automated Pet Feeder System
- Greenhouse Monitoring System
- Vertiv NetSure™ Control Unit (NCU) – Product Datasheet

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LANGUAGES

English | Hindi | Marathi